

YONG JIN (JINA) CHUN

Ph.D. Student, University of California, Irvine ✉ chunyj@uci.edu

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SUMMARY

Yong Jin (Jina) Chun is a PhD student in Software Engineering at the University of California, Irvine, advised by Professor Iftekhar Ahmed. She completed a Master's degree in Software Engineering from the University of California, Irvine. Her research advances software engineering through the use of Large Language Models (LLMs), Natural Language Processing (NLP), and Multi-Agent Systems. She focuses on developing and evaluating collaborative, debate-driven agentic frameworks to improve reliability, reasoning, and scalability in automated software development. She also serves as a reviewer for top-tier journals such as IEEE Transactions on Reliability, ACM Transactions on Software Engineering and Methodology (TOSEM), and Empirical Software Engineering (EMSE).

EDUCATION

PhD in Software Engineering , University of California, Irvine Advisor: Professor Iftekhar Ahmed	<i>Sept 2022 - Current</i> GPA: 4.0/4.0
MS in Software Engineering University of California, Irvine Advisor : Professor Iftekhar Ahmed Dissertation: Multi-Agent Debate for Advancing Agentic Software Development	<i>Sept 2022 - Sept 2025</i> GPA: 4.0/4.0
BS in Computer Science Boston University Cum Laude, Dean's Honor List	<i>Aug 2017 - May 2021</i> GPA: 3.6/4.0

WORK EXPERIENCES

Research Intern , SAINTES Group, Advisor: Christian Bird	<i>June 2026 – Sept 2026 (Present)</i> Redmond, Washington
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- Designed a working-knowledge memory system that integrates software engineering artifacts, organizational communications, and AI agent traces to provide contextual retrieval across the development lifecycle.
- Investigated memory representations, retrieval strategies, and privacy-aware sharing mechanisms to improve developer productivity, onboarding, code review, and agent-assisted software engineering workflows.

Research Assistant , STAIRS lab	<i>Sept 2022 – Current</i> University of California, Irvine
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- Published two peer-reviewed papers on Large Language Models (LLMs) in software engineering.
- Developed multi-agent debate frameworks to analyze limitations and enhance the performance of LLM-driven software engineering tasks.
- Conducted an empirical analysis of prompting techniques across diverse software engineering tasks to investigate factors influencing their effectiveness.

Data Science Intern , Orion Health	<i>Jan 2020 – Apr 2020</i> Auckland, New Zealand
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- Analyzed data annotation strategies to enhance accuracy and efficiency in clinical data processing.
- Developed a Natural Language Processing (NLP)-based annotation tool for clinical data.

Software Engineering Intern,
Parasoft Southeast Asia

Jun 2019 – Aug 2019
Singapore

- Investigated the company's automated software testing product to identify software quality issues.
- Designed and developed a user interface improvement based on the findings.

PUBLICATIONS

Jina Chun, Iftekhar Ahmed (2026). What Do Agents Communicate? Characterizing Information Exchange in Multi-Agent Systems. Manuscript submitted for publication in the *Conference on Neural Information Processing Systems (NeurIPS)*.

Jina Chun, Qihong Chen, Jiawei Li, Iftekhar Ahmed (2026). Enhancing LLM Performance Through Debate: An Empirical Study on Multi-Agent Debate for Software Engineering Tasks. *ACM Transactions on Software Engineering and Methodology (TOSEM)*.

EG Santana Jr, Gabriel Benjamin, **Jina Chun**, Melissa Araujo, Harrison Santos, David Freitas, Eduardo Almeida, Paulo Anselmo da Neto, Jiawei Li, Iftekhar Ahmed (2025). Which Prompt Works Best? An Empirical Evaluation of Prompting Techniques For Software Engineering Tasks. (*Arxiv*)

SKILLS

Research Skills

Natural Language Processing, Large Language Models, Multi-Agent Systems, Agentic Memory Systems, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Context Engineering, Developer Productivity Tools, Software Development Lifecycle (SDLC), Fine-tuning, Software Testing (JUnit, PyTest), Prompt Engineering, Model Training, Machine Learning, Dataset Curation and Labeling, Literature Review, Empirical Research, Qualitative and Statistical Analysis, Human Subjects Research

Programming Languages and Frameworks

Java, Python, JavaScript, TypeScript, Node.js, HTML, CSS, Django, React, Bootstrap

Machine Learning and AI Frameworks

PyTorch, TensorFlow, LangChain, scikit-learn, Hugging Face Transformers, OpenAI API

Data Analysis

Pandas, NumPy, Matplotlib, Seaborn, SQL, Data Visualization

Developer Tools and Platforms

Git, GitHub, Docker, Heroku, Linux, VS Code, Jupyter Notebook, Conda, Bash

PROJECTS

AI-Driven Mental Health Counseling

- Investigated whether open-source LLMs can replicate human counselor linguistic patterns.
- Applied various Natural Processing Techniques to compare the AI and human-generated counseling responses.

Culture Explorer: Responsive Web Platform for Cultural Discovery

- Developed a responsive website showcasing global destinations and cultural experiences.
- Designed and deployed interactive gallery and destination pages optimized for responsiveness, usability, and cross-browser performance.

DineLog: Interactive Food Review and Sharing Web Platform

- Built a full-stack Django application supporting user registration, page creation, quote posting, and media uploads.
- Implemented modular apps with authentication, CRUD functionality, and Heroku-ready configuration. (Demonstration Video)

PROFESSIONAL SERVICE

Reviewer , IEEE Transactions on Reliability	<i>April 2025</i>
Reviewer , ACM Transactions on Software Engineering and Methodology (TOSEM)	<i>April 2024</i>
Reviewer , ACM Empirical Software Engineering	<i>July 2025</i>

AWARDS AND HONORS

Chair's Award *Sept 2023*
Donald Bren School of Information and Computer Sciences, University of California, Irvine
 Received upon admission to the Ph.D. program and granted in recognition of exceptional potential to make significant research contributions to the Informatics community.

Cum Laude *June 2021*
Boston University, College of Arts & Sciences
 Rewarded upon graduation in recognition of consistent academic excellence and distinguished achievement throughout undergraduate study.

Dean's List *August 2018- June 2021*
Boston University, College of Arts & Sciences
 Received each semester throughout four years of study in recognition of consistent academic excellence and outstanding scholastic achievement.

STUDENT MENTOR

Mentor, ICS Honors Program University of California, Irvine	<i>July 2023 - Ongoing</i>
Mentor, Graduate International Connection Program University of California, Irvine	<i>March 2023 - Ongoing</i>
International Peer Mentor Program Boston University	<i>Aug 2017 - June 2021</i>